

are also called bi-implications.

Table 6: The Truth Table for the Biconditional $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

$(p \rightarrow q)$	$(q \rightarrow p)$
T	T
F	F
F	F
T	T

Common ways to express $p \leftrightarrow q$

- "p is necessary and sufficient for q"
- "if p then q, and conversely"
- "p if and only if q"
- "p exactly when q"

$(p \rightarrow q) \wedge (q \rightarrow p)$

$(p \rightarrow q)$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent

Example 14: Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$

p	q	$(p \wedge q)$	$\neg q$	$(p \vee \neg q)$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	T	F	T	F
T	F	F	T	T	F
F	T	F	F	F	T
F	F	F	T	T	F

Table 8: Precedence of Logical Operators

Operators	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

A bit is a symbol with two possible values, namely, 0 (zero) and 1 (one). This meaning of the word bit comes from binary digit, because zeros and ones are the digits used in binary representation of numbers.

A variable is called a Boolean variable if its value is either true or false. Consequently, a Boolean variable can be represented using a bit.

Truth value	Bit
T	1
F	0